

Problem

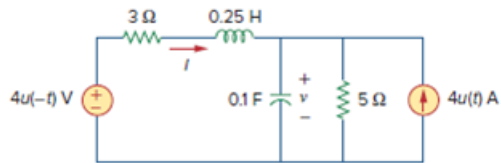
In the circuit of Fig. 8.65, find:

(a) $v(0^+)$ and $i(0^+)$,

(b) $dv(0^+)/dt$ and $di(0^+)/dt$,

(c) $v(\infty)$ and $i(\infty)$.

Figure 8.65:



Step-by-step solution

Step 1 of 10

Draw the circuit diagram:

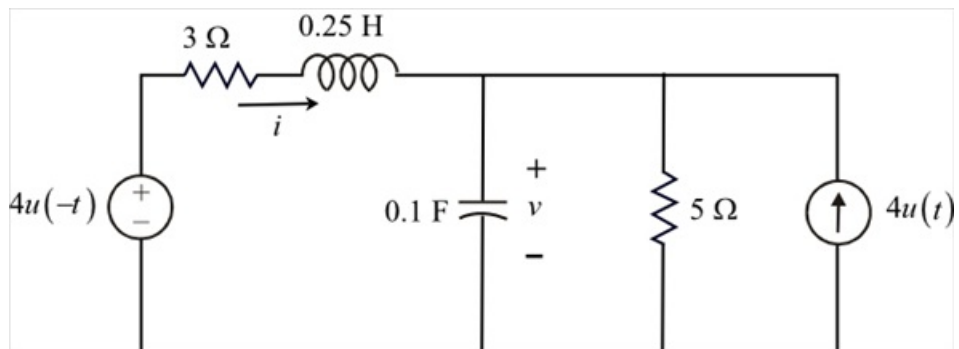


Figure 1

Step 2 of 10

(a)

Calculate the source values for $t < 0$.

$$\begin{aligned}4u(-t) &= (4)(1) \\ &= 4 \text{ V}\end{aligned}$$

$$\begin{aligned}4u(t) &= (4)(0) \\ &= 0 \text{ A}\end{aligned}$$

Initially at $t < 0$, the capacitor acts like an open circuit while the inductor acts like a short circuit. The equivalent circuit is shown Figure 2.

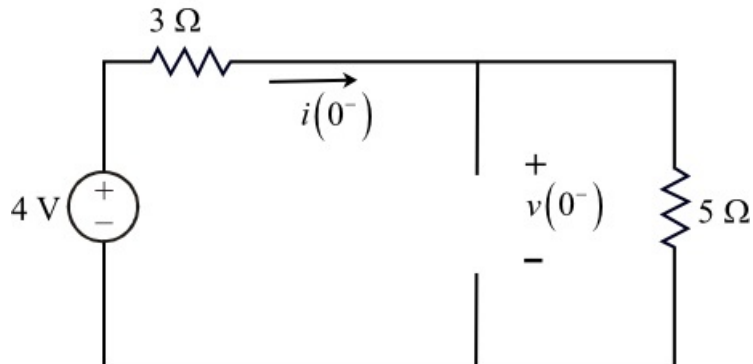


Figure 2

Step 3 of 10

Calculate the initial current, using Ohm's law.

$$\begin{aligned}i(0^-) &= \frac{4}{3+5} \\ &= 0.5 \text{ A}\end{aligned}$$

Calculate the initial voltage using Ohm's law.

$$\begin{aligned}v(0^-) &= (5)(0.5) \\ &= 2.5 \text{ V}\end{aligned}$$

Hence capacitor voltage $v(0^-) = 2.5 \text{ V}$ and inductor current $i(0^-) = 0.5 \text{ A}$ does not change abruptly, hence at $t > 0$ the value remains same as $v(0^-) = v(0^+) = 2.5 \text{ V}$ and

$$i(0^-) = i(0^+) = 0.5 \text{ A}.$$

Step 4 of 10

(b)

Calculate the source values for $t = 0^+$.

$$\begin{aligned}4u(-t) &= (4)(0) \\ &= 0 \text{ V}\end{aligned}$$

$$\begin{aligned}4u(t) &= (4)(1) \\ &= 4 \text{ V}\end{aligned}$$

Draw the circuit diagram at $t = 0^+$.

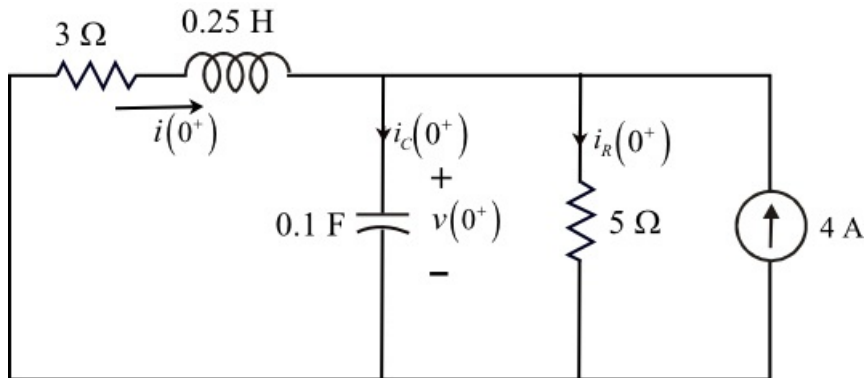


Figure 3

Step 5 of 10

Calculate the current through 5Ω resistor.

$$\begin{aligned}i_R(0^+) &= \frac{v(0^+)}{5} \\ &= \frac{2.5}{5} \\ &= 0.5 \text{ A}\end{aligned}$$

Step 6 of 10

Calculate the current through capacitor using Kirchhoff's current law.

$$\begin{aligned}i_C(0^+) + i_R(0^+) &= i(0^+) + 4 \\ i_C(0^+) + 0.5 &= 0.5 + 4 \\ i_C(0^+) &= 4 \text{ A}\end{aligned}$$

Step 7 of 10

Write the expression for capacitor current at $t = 0^+$.

$$\begin{aligned}i_C(0^+) &= C \frac{dv(0^+)}{dt} \\ \frac{dv(0^+)}{dt} &= \frac{i_C(0^+)}{C} \\ &= \frac{4}{0.1} \\ &= 40 \text{ V/s}\end{aligned}$$

Hence, the value of $\frac{dv(0^+)}{dt}$ is 40 V/s.

Apply Kirchhoff's voltage law to left side loop in Figure 3.

$$3i(0^+) + v_L(0^+) + v(0^+) = 0$$

$$3(0.5) + v_L(0^+) + 2.5 = 0$$

$$1.5 + v_L(0^+) + 2.5 = 0$$

$$v_L(0^+) = -4 \text{ V}$$

Write the expression for inductor voltage at $t = 0^+$.

$$v_L(0^+) = L \frac{di(0^+)}{dt}$$

$$\frac{di(0^+)}{dt} = \frac{v_L(0^+)}{L}$$

$$= \frac{-4}{0.25}$$

$$= -16 \text{ A/s}$$

Hence, the value of $\frac{di(0^+)}{dt}$ is $\boxed{-16 \text{ A/s}}$.

Step 9 of 10

(c)

Draw the modified circuit of Figure 1 at $t = \infty$.

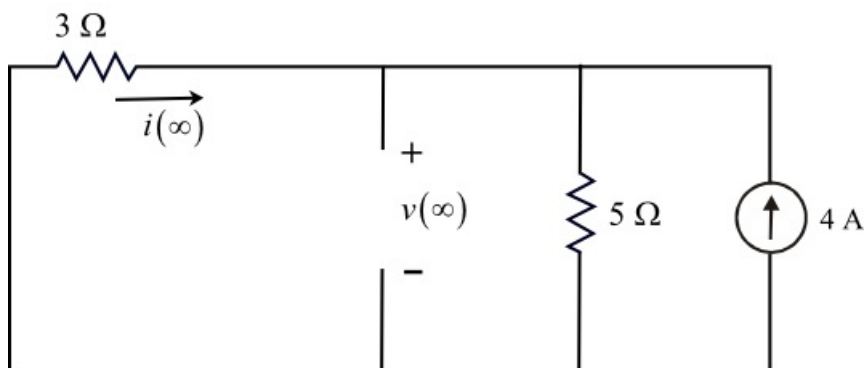


Figure 4

Step 10 of 10

Apply current divider rule to Figure 4, to obtain the current $i(\infty)$.

$$\begin{aligned}i(\infty) &= (-4) \left(\frac{5}{3+5} \right) \\&= (-4) \left(\frac{5}{8} \right) \\&= -2.5 \text{ A}\end{aligned}$$

Hence the final value of inductor current, $i(\infty)$ is $\boxed{-2.5 \text{ A}}$.

Calculate the final value of the capacitor voltage $v(\infty)$.

$$\begin{aligned}v(\infty) &= (3)(-i(\infty)) \\&= (3)(-(-2.5)) \\&= 7.5 \text{ V}\end{aligned}$$

Hence the final value of capacitor voltage, $v(\infty)$ is $\boxed{7.5 \text{ V}}$.